Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/22/2016
PI Name:	Karma, Alain Ph.D.		
Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	a.karma@neu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617 373 2929
Organization Name:	Northeastern University		
PI Address 1:	Department of Physics		
PI Address 2:	360 Huntington Ave		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	11/16/2015	**End Date:**	11/15/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Lee, Jonathan	**Contact Phone:**	
Contact Email:	jonathan.a.lee@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AB54G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR's NOTE (12/14/2015): Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX14AB34G." Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. The production of innovative materials and the optimization of both properties and processing routes go through an accurate prediction and modeling of the dynamics of microstructures formation. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity experiments provide		

Task Description:	a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. The main goal of this project is to develop state-of-the-art computational tools to model dendritic microstructures formation under diffusive growth conditions and to establish benchmark comparisons with experimental observations under microgravity conditions. Those comparisons have the dual purpose to obtain new fundamental insights into microstructure formation and to validate computational models that can ultimately be used in an industrial setting to guide the design of structural alloys and optimize solidification processing conditions. Computational tools include the phase-field method and a new dendritic needle network (DNN) model. The phase-field method resolves the details of the dendrite tip scale and is ideally suited to model cellular and dendritic structures in dilute alloys on a scale comparable to the primary spacing. In the framework of a collaboration between NASA and the French National Space Agency (CNES), phase-field simulations are being used to model directional solidification experiments in a transparent organic succinonitrile–camphor alloy under low gravity in the Device for the Study of Critical Liquids and Crystallization (DECLIC) on-board the International Space Station (ISS). The DNN model in turn is designed to model dendritic microstructure formation on a grain scale that is orders of magnitude larger than the dendrite tip scale. Unlike previous models of grain structure evolution, the DNN approach models explicitly the growth kinetics of all branches of the hierarchical dendritic network (including primary, secondary, and tertiary branches) that interact diffusively via the solutal field. Hence, this method is ideally suited to investigate complex mechanisms of dendritic microstructure formation including the competitive growth of misoriented columnar grains and the columnar-to-equiaxed transition. This transition is of practical relevance for the casting industry. It is being investigated experimentally aboard the ISS in collaboration with our European partners in the framework of the CETSOL project. Simulation results are critically important to interpret the results of those experiments in both metallic and transparent organic alloys and to guide future experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to lightweight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	During the current funding period, we made major progress in further developing the DNN approach to model the columnar-to-equiaxed transition (CET) that is being investigated in metallic Al-based alloys in the framework of the CETSOL project. We also used our established quantitative phase-field modeling approach to investigate the mechanisms of grain selection in polycrystalline samples and to model the cellular/dendritic microstructures being observed in situ in transparent organic alloys in DECLIC/DSI. In the framework of the CETSOL project, we extended the DNN model to model the growth of randomly oriented grains in three dimensions and to incorporate experimental measurements of the temperature field at different locations along the sample. Those extensions of the model are important to quantitatively model experimental observations of the columnar-to-equiaxed transition in Al-7wt%Si samples solidified aboard the ISS. In addition, we performed a comprehensive phase-field study of columnar grain growth competition during directional solidification of bi-crystalline samples. This study was motivated by our previous finding that the competition between columnar and elongated equiaxed grains leads to progressive columnar-to-equiaxed transitions that have been observed in Al-7wt%Si samples. Our study shows that grain-boundary orientation selection depends strongly on whether the primary dendrite growth directions in the two grains have lateral components perpendicular to the temperature gradient axis that point in the same or opposite directions. Based on an exhaustive survey of different grain orientations, we have identified a range of orientations in which grain selection follows the classical description of Walton and Chalmers and another range in which selection is predominantly controlled by unusual overgrowth of favorably-oriented dendrites at a converging grain boundary. Based on those results, we developed a simple analytical model to predict grain selection. This model shows a remarkably good quantitative agreement with both simulations and experiments. In the framework of the DECLIC project, we have carried out three-dimensional phase-field simulations to model unique experimental observations of cellular and dendritic array structures formed under diffusive growth conditions aboard the ISS. We investigated two important new effects on the dynamical selection of the cellular/dendritic array spacing. The first is a recently characterized drift of isotherms associated with the transient initial recoil of the solid-liquid interface. This recoil is characterized by an initial acceleration of the interface before it reaches a steady-state regime where the growth velocity is equal to the external pulling velocity. Simulations reveal that this thermal drift leads to a substantial increase of the dynamically selected primary spacing, thereby improving the quantitative agreement between simulated and observed spacings. Secondly, we investigated the effect of a finite misorientation between the principal crystal axes and the temperature gradient axis. Phase-field simulations show that a small misorientation of only a few degrees can have a profound influence on microstructure selection. In particular, we have identified a novel propagative “source-sink” mechanism of spacing selection that operates on longer time scales than the traditional competitive growth mechanism acting during the transient recoil of the interface. We find that tertiary branching at the diverging edge of the sample acts as a “source” of new cells that propagate a larger uniform spacing inside the sample, while the converging edge of the sample acts as a “sink” where cells are eliminated. Based on those insights, we developed an analytical model to predict the dynamically selected spacing on long time scales. This fundamental knowledge is key to improving textured polycrystals for advanced materials. The next step in the CETSOL project is to apply the DNN approach to model quantitatively the columnar-to-equiaxed transition in three dimensions in refined and non-refined Al-7wt%Si samples processed under different conditions aboard the ISS. Simulation results will be compared to detailed measurements of grain size and orientation in those

	samples using serial sectioning. In addition, the DNN model will be used to investigate the growth of equiaxed dendrites solidified in a microgravity environment in the transparent model alloy Neopentylglycol-dCamphor during the sounding rocket mission MASER-13. The next step in the DECLIC project is to model in situ observations of the effects of crystal misorientation, grain boundary configurations, and macroscopic interface curvature on microstructure and spacing selection in bulk samples in the transparent succinonitrile-camphor system.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Articles in Peer-reviewed Journals	Tourret D, Karma A. "Three-dimensional dendritic needle network model for alloy solidification." Acta Materialia. 2016 Nov;120:240-54. Available online 1 September 2016. http://dx.doi.org/10.1016/j.actamat.2016.08.041 , Nov-2016
Articles in Peer-reviewed Journals	Karma A, Tourret D. "Atomistic to continuum modeling of solidification microstructures." Current Opinion in Solid State and Materials Science. 2016 Feb;20(1):25-36. http://dx.doi.org/10.1016/j.cossms.2015.09.001 , Feb-2016
Articles in Peer-reviewed Journals	Mota FL, Fabietti LM, Bergeon N, Strutzenberg LL, Karma A, Billia B, Trivedi R. "Quantitative determination of the solidus line in the dilute limit of succinonitrile–camphor alloys." Journal of Crystal Growth. 2016 Aug 1;447:31–5. http://dx.doi.org/10.1016/j.jcrysgro.2016.04.054 , Aug-2016
Awards	Karma A. "John Hunt Medal from the Institute of Materials, Minerals and Mining (IOM3), April 2016." Apr-2016